CS315 · ALGORITHMS ANALYSIS & DESIGN

Practice 4 Practice

DP, greedy and B-trees · Lectures 4 to 7

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01 Questions

Q1 TRUE / FALSE

1. Dynamic programming works when a problem has optimal sub-structure and overlapping sub-problems. ( )
2. The recursive Fibonacci algorithm runs in O(n). ( )
3. A greedy algorithm may reverse an earlier decision if it turns out to be wrong. ( )
4. Greedy always gives the optimal solution for the 0-1 knapsack problem. ( )
5. Greedy gives the optimal solution for the fractional knapsack problem. ( )
6. In a Huffman code, frequent characters get shorter codes. ( )
7. All leaves of a B-tree are at the same depth. ( )
8. In a B-tree with minimum degree t, every non-root node has at most 2t keys. ( )
9. A B-tree grows in height when its root is split. ( )
10. B-trees are designed mainly for data stored in RAM. ( )

Q2 MCQ

1. The cost of multiplying a 10×20 matrix by a 20×5 matrix is: a) 35 b) 200 c) 1000 d) 4000

2. The running time of the greedy activity selector on activities sorted by finish time is: a) $\Theta(n)$ b) $\Theta(n \log n)$ c) $\Theta(n^2)$ d) $\Theta(2^n)$

3. The time complexity of Huffman coding is: a) $O(n)$ b) $O(n \log n)$ c) $O(n^2)$ d) $O(\log n)$

4. In a B-tree with $t = 3$, a non-root node holds between: a) 1 and 3 keys b) 2 and 5 keys c) 3 and 6 keys d) 2 and 6 keys

5. Activity selection greedily picks the activity with the earliest: a) start time b) finish time c) duration d) index

6. Deleting a key from an internal node whose two neighbouring children both have $t - 1$ keys uses: a) Case 1 b) Case 2a c) Case 2c (merge) d) Case 3a

Q3 MATRIX CHAIN

Find the minimum number of scalar multiplications and the optimal parenthesization for A = 30×35, B = 35×15, C = 15×5, D = 5×10.

	A	B	C	D
A				
B				
C				
D				

Q4 LCS

Using dynamic programming, find the length of the LCS of X = BACDB and Y = BDCB, and give one LCS.

	⌀	B	D	C	B
⌀					
B					
A					
C					
D					
B					

Q5 ACTIVITY SELECTION

Select the maximum set of compatible activities:

a	1	2	3	4	5	6	7
s	1	2	4	1	5	8	9
f	3	5	7	8	9	10	11

Q6 KNAPSACK

W = 10. Items: (w = 5, v = 30), (w = 4, v = 40), (w = 6, v = 30), (w = 3, v = 6).

a) Solve the fractional knapsack with the greedy method.

b) What does greedy give if items must be taken whole (0-1)? Is it optimal?

Q7 HUFFMAN

Build the Huffman tree and codes for the frequencies: a = 45, b = 13, c = 12, d = 16, e = 9, f = 5. How many bits does the encoded text need, and how many with a fixed 3-bit code?

Q8 B-TREE INSERT

Insert 10, 20, 5, 6, 12, 30, 7, 17 into an empty B-tree with minimum degree t = 2 (split full nodes on the way down). Draw the final tree.

Q9

B-TREE DELETE

In the slides' tree (t = 3) below, delete C. Which case applies? Draw the result.

